EXP_2026Q2_SEARCH_RELEVANCE · VERDICT REPORT · ENGINEER

SEARCH · RELEVANCE · 2026-06-05 · 12,000 UNITS · 14 DAYS

Do not ship.

Primary lift may be real, but at least one guardrail's lower 90% CI crosses the harm threshold.

TL;DR

DO NOT SHIP. **Hold the change. The primary moved but the guardrail breached.**

search_to_purchase_rate lifted by +0.99pp, but the revenue-per-user guardrail's 90% CI fell past its harm threshold. The compositional story (more orders, lower value) shipped is a net negative.

GUARDRAIL · REVENUE_PER_USER

-19.26%

treatment vs control — 90% CI past the −5% harm threshold

90% CI ON GUARDRAIL

[-24.26, -14.26]%

HARM THRESHOLD

-5.0%

PRIMARY LIFT

+0.99pp

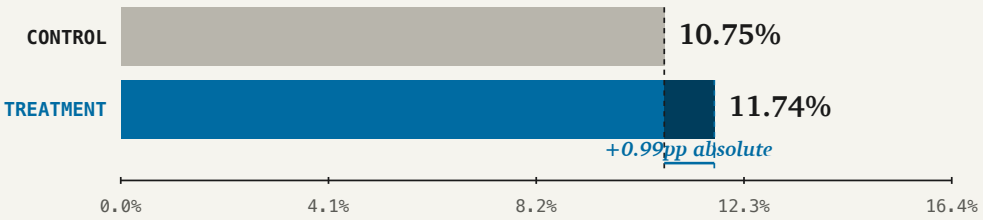
DECISION STEP

step 2

PRIMARY METRIC · SEARCH_TO_PURCHASE_RATE

Treatment 11.74% vs Control 10.75%

Treatment and control rates on the same axis. The dark blue segment on the treatment bar is the absolute lift; the dotted line shows the lift relative to the control end.

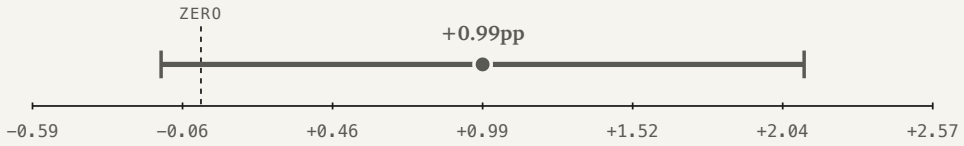


Per-arm conversion. The lift is +0.99pp absolute (+9.2% relative).

CONFIDENCE INTERVAL ON LIFT

95% CI on lift: (-0.14, +2.12)pp — straddles zero.

Zero is inside the interval. We can't reject the null at a = 0.05. The verdict tree then asks whether we had enough sample to detect what we cared about.



The CI bar shows the range of plausible true lifts. The crossbars are the 95% bounds; the dot is the point estimate.

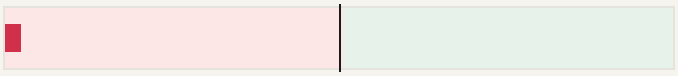
GUARDRAILS

1 guardrail breached.

At least one guardrail's lower 90% CI fell past its harm threshold. The verdict tree fired NO-SHIP-GUARDRAIL at step 2 — primary metric movement does not outweigh a guardrail breach.

revenue_per_user

`threshold ≥ −5.0% · higher_is_better`



`90% CI [−24.26, −14.26]% · realized −19.26% ·` BREACH

DECISION TREE · HOW THE VERDICT WAS REACHED

Walked 8 steps; step 2 fired the verdict.

Each row is a checkpoint in the verdict tree. Walking stops at the row marked FIRED; that step's logic is what produced the verdict above.

0	Required inputs present (UNVERIFIABLE guard)	`all non-null`
1	SRM passes (or override resolved)	`srm_pass = True`
2	Guardrails clear (no 90% CI on harm side) · **FIRED**	`guard_lift = -19.26%`
3	Powered (n_observed ≥ n_required)	`n = 12,000 / 9,600`
4	Primary CI excludes 0	`CI [-0.14, +2.12] pp`
5	Primary CI not on harm side	`direction = higher_is_better`
6	Lift magnitude ≥ MDE/2	`|lift| = 0.990pp vs MDE/2 = 0.100pp`
7	Novelty stable (late_ratio ≥ 0.7)	`late_ratio = —`

PROVENANCE · AUDITABLE CHAIN

Reproducible from sealed brief + warehouse fixture.

Every quantitative claim above resolves to a row in the warehouse, a function in `agentxp.stats.*`, or a step in `agentxp.interpret.tree`. The chain hash below covers the brief, every stat result, and the verdict tree's terminal step.

```
EXPERIMENT_ID          exp_2026q2_search_relevance
CHAIN_HASH
087533a307a629f075c038be8dabd30fe92f9c0252f2f4651f038e71a7cafe91
RENDER_STATUS          VERIFIED
AGENTXP_VERSION        0.3.0-v3
GENERATED_AT           2026-06-05T15:42:15Z
PRIMARY_METRIC         search_to_purchase_rate · proportion_test
SRM_TEST               srm_chi_squared · threshold 0.0005
TREE_MODULE            agentxp.interpret.tree.walk_tree
REPRODUCE              python scratch/walk_experiment.py
exp_2026q2_search_relevance
```